Applied-Qual-Fall-2024

October 7, 2024

1 Instructions

- **This is a take-home exam**. Attempt all 6 problems; submit your answers by editing them into this jupyter notebook, running them, and emailing the completed notebook to Benjamin Young by the deadline (48 hours after the exam was made available).
- **This is an open-book exam**. You may use whatever reference materials you wish, including existing materials on the internet, your textbooks, previous problem sets, and your notes.
- **Please cite your sources** in your responses to each problem.
- **Please work on your own** and please don't post the exam online or otherwise ask for help on these problems.
- **Please explain your answers clearly in a markdown cell, in addition to any code you write**. Before handing in your code, please run it yourself - ideally, the graders of this exam shouldn't need to rerun your code (we *may* do that, in an attempt to figure out what you've done or some such thing, but we shouldn't *need* to).

2 Problem 1 - Airy Equation

This ODE is known as the Airy equation:

$$\frac{d^2y}{dx^2} - xy = 0.$$

One of the solutions is the Airy function Ai(x), which has the property that

$$\lim_{x\to\infty} Ai(x) = 0.$$

If we also specify values $y(0), y'(0)$, then we have an initial value problem. Write code to solve this initial value problem numerically, using an appropriate finite-difference technique; your solution should take in values for $y(0), y'(0)$, a step size, and a maximum x value, x_{max}; it will produce a list of values for a function y which approximately solves the DE on the interval $[0, x_{\text{max}}]$. Plot your solution when $y(0) = 1, y'(0) = 0$.

Describe the error analysis of the method you have implemented.

I imagine that none of the numerical solutions you find will have the above asymptotic property, even if you use initial values that are close to those of $Ai(x)$ - rather, I imagine that the solution will almost always blow up as x gets large. Explain why.

3 Problem 2 - Wave Equation

This problem asks you to solve the wave equation in one spatial dimension:

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial t^2}$$

on the time domain $[0, 2\pi]$ and space domain $[0, 2\pi]$. Evidently, one solution to this equation is

$$U(t, x) = \sin(t)\sin(x).$$

Verify that knowing - $u(0, x) = U(0, x)$ and $u_t(0, x) = U_t(0, x)$ for $0 \leq x \leq 2\pi$, - $u(t, 0) = U(t, 0)$ and $u(t, 2\pi) = U(t, 2\pi)$ for $0 \leq t \leq 2\pi$

is sufficient information to approximately reproduce the solution $u = U$ numerically. Then, solve this IVP numerically; compare your solution to the exact solution. Use any appropriate method.

4 Problem 3 - Call Center Statistics

Suppose that a call center expects to receive on average λ calls per minute. Assuming that the timing of these calls is random and uncorrelated, the number of calls k received per minute, where k is a nonnegative integer, can then be modeled by the probability distribution

$$p(k) = \frac{\lambda^k e^{-k}}{k!}.$$

Exercise

(a) Given a set of observations $\{k_1, \ldots, k_N\}$, compute the maximum-likelihood estimate of the parameter λ.

(b) Use Stirling's approximation and show that we can write the distribution as $p(k) = e^{f(k)}$. What is $f(k)$?

(c) Interpreting $k \in \mathbb{R}$ as a continuous variable, perform a Taylor expansion of $f(k)$ to second order around its maximum to show that we can approximate $p(k)$ as a normal distribution (don't worry about the normalization prefactor). Report the mean and variance of the normal distibution.

(d) In which limit does $p(k)$ become (if we allow $k \in \mathbb{R}$ rather than $k \in \mathbb{Z}$) exactly equal to a normal distribution? Explain why in detail.

5 Problem 4 - Mixture of exponential distributions

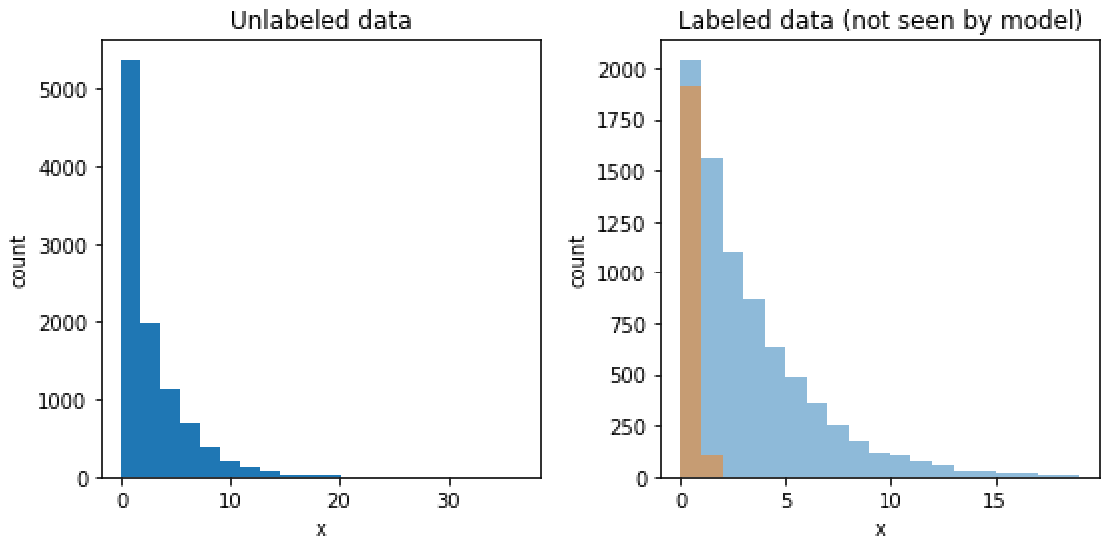
In the block of code below, we generate data from two exponential distributions: $x_n \sim p(x)$, where

$$p_\alpha(x) = \begin{cases} \lambda_\alpha e^{-\lambda_\alpha x}, x \geq 0 \\ 0, \quad \text{else} \end{cases}$$

where $\lambda_\alpha > 0$, and $\alpha \in \{1, 2\}$.

[4]:

```python
import numpy as np
import matplotlib.pyplot as plt
# Make some fake data.

n = 10000  # total number of data points

# Parameters for the two distributions
# (we'll use underscores to denote the true parameters, which we'll try to
 ↪infer below):
lambda1_ = 0.3
lambda2_ = 3

# The mixture probabilities:
pi1_, pi2_ = 0.8, 0.2

# For each data point, choose a cluster that it belongs to and generate its
 ↪location:
x = []
labels = []
for ii in range(n):
    if np.random.rand() < pi1_:
        x.append(np.random.exponential(1/lambda1_))
        labels.append(0)
    else:
        x.append(np.random.exponential(1/lambda2_))
        labels.append(1)

# Plot the data:
plt.figure(figsize=(8,4))
plt.subplot(121)
plt.title('Unlabeled data')
plt.hist(x, bins=20)
plt.xlabel('x')
plt.ylabel('count')

plt.subplot(122)
plt.title('Labeled data (not seen by model)')
plt.hist(np.array(x)[np.array(labels)==0], alpha=0.5, bins=np.arange(20))
plt.hist(np.array(x)[np.array(labels)==1], alpha=0.5, bins=np.arange(20))
plt.xlabel('x')
plt.ylabel('count')
plt.tight_layout()
```



In this exercise, given the unlabeled data, we will use the Expectation Maximization algorithm to try to recover the statistical model that was used to generate this data, inferring the parameters $\Theta = \{\lambda_1, \lambda_2, \pi_1, \pi_2\}$.

Recall that the EM algorithm can be summarized as follows:

1. Choose initial guess for parameters Θ^{old}.
2. Iterate:

   2a. E step: Evaluate $p(\mathbf{Z}|\mathbf{X}, \Theta^{\text{old}})$. Concretely, this consists of using Θ^{old} to calculate the *responsibilities* $\gamma(z_{nk})$, where $z_{nk} \in \{0, 1\}$, n is the data point, and k is the latent-variable index.

   2b. M step: Using the responsibilities calculated in the E step, let $\Theta^{\text{new}} = \text{argmax}_\Theta Q(\Theta, \Theta^{\text{old}})$, where $Q(\Theta, \Theta^{\text{old}}) = \sum_{\mathbf{Z}} p(\mathbf{Z}|\mathbf{X}, \Theta^{\text{old}}) \ln p(\mathbf{X}, \mathbf{Z}|\Theta)$.

   2c. Check for convergence of the log-likelihood and/or Θ. If not converged, update $\Theta^{\text{old}} \leftarrow \Theta^{\text{new}}$.

Exercise

(a) Give the mathematical expressions for the responsibilities $\gamma(z_{nk})$ and the update equations for λ_α in terms of $\pi_{1,2}$, $\lambda_{1,2}$, and the data $\{x_n\}_{n=0}^{N}$.

(b) Implement the EM algorithm to find the maximum likelihood solution for a mixture of exponentials model with $K = 2$ applied to the data above. Report the inferred values of the parameters Θ. Plot the value of the log likelihood at each step in the iteration and show that it converges. The algorithm should be implemented from scratch, with only basic Numpy functions being used.

6 Problem 5 - Lattice Paths

Let S be the set of lattice paths from (0,0) to (20, 20), taking unit-length steps up and to the right. Consider the probability measure π on S such that the probability of the path P is proportional to $0.9^{A(P)}$, where $A(P)$ is the area between P and the x axis.

Describe a markov chain whose stationary distribution is π. Then use monotome coupling fr|om the past to draw a sample from π.

7 Problem 6 - Poisson point processes

This problem has two parts: *random sampling* and *inference*.

sampling: Let $f(x) = 100e^{-x^2}$. Write a program to sample a data set S from a Poisson point process on $[-5, 5]$ with intensity $f(x)$.

inference: Given such an S, as well as the information that S is a sample from a Poisson point process with some intensity g, describe how you might infer what g was. Do this for your set S, and plot the true density f and the inferred density g on the same axes.